

NON-REAL-ROOTED h^* -VECTORS OF ORDER POLYTOPES FROM PERMUTATION POSETS

PER ALEXANDERSSON

ABSTRACT. We investigate the real-rootedness of h^* -polynomials of order polytopes arising from permutation posets. Starting from a permutation of Stembridge whose weak-order descent polynomial fails real-rootedness, we observe that the associated *permutation poset* yields an order polytope whose h^* -polynomial (equivalently, the P -Eulerian polynomial) is also not real-rooted. A computational search over 34,226 nearby 4321-avoiding permutations finds exactly 22 non-real-rooted examples, all of which are log-concave.

1. SETUP

Given a permutation $w = (w_1, \dots, w_n) \in S_n$, the *permutation poset* P_w is the partial order on $\{1, \dots, n\}$ defined by

$$i <_{P_w} j \iff i < j \text{ and } w_i < w_j.$$

This is always naturally labeled (every cover relation $i < j$ has $i < j$ as integers), so by Stanley's theorem [Sta11, Thm. 4.5.14] the h^* -vector of the order polytope $\mathcal{O}(P_w)$ equals the P -Eulerian polynomial:

$$h_k^* = \#\{\sigma \in \mathcal{L}(P_w) : \text{des}(\sigma) = k\},$$

where $\mathcal{L}(P_w)$ is the set of linear extensions and $\text{des}(\sigma) = \#\{i : \sigma(i) > \sigma(i+1)\}$.

If w avoids the pattern 321, then the longest decreasing subsequence of w has length ≤ 2 , so by Dilworth's theorem P_w has width ≤ 2 . This makes both the frontier DP (for Ehrhart computation) and linear extension enumeration (for direct P -Eulerian computation) very efficient.

2. STEMBRIDGE'S PERMUTATION

Consider the 321-avoiding permutation on 17 elements:

$$w_0 = (2, 4, 6, 8, 10, 1, 12, 3, 15, 5, 17, 7, 9, 11, 13, 14, 16).$$

This permutation was identified by Stembridge in connection with weak-order descent polynomials. The permutation poset P_{w_0} has 25 cover relations,

Date: March 27, 2026.

7,016 linear extensions, and h^* -polynomial

$$h^*(t) = 1 + 32t + 336t^2 + 1420t^3 + 2534t^4 + 1946t^5 + 658t^6 + 86t^7 + 3t^8.$$

This polynomial is **log-concave** but **not real-rooted**.

3. NEIGHBORHOOD SEARCH

Starting from w_0 , we generate all permutations reachable by at most d arbitrary transpositions (not just adjacent), filter to those avoiding the pattern 4321, and test the h^* -polynomial of each permutation poset for real-rootedness.

The h^* -polynomial is computed via the P -Eulerian polynomial (streaming linear extension enumeration with descent counting, using $O(n)$ memory). For the 321-avoiding permutations in this family, the number of linear extensions is moderate ($\leq 16,000$), so this is fast (34,226 posets tested in 14 seconds using parallel computation).

TABLE 1. BFS neighborhood of w_0 by transposition distance d .

d	BFS total	4321-avoiding	not real-rooted	rate
0	1	1	1	100%
1	136	65	5	7.7%
2	8,500	1,850	8	0.43%
3	323,680	32,310	8	0.025%
total	332,317	34,226	22	0.064%

Observation 3.1. Non-real-rootedness is rare: only 22 out of 34,226 permutations fail, and the rate drops sharply with distance. All 22 failures are log-concave. Of the 22, exactly one contains the pattern 321 (at distance 3); the remaining 21 are 321-avoiding.

4. THE 22 NON-REAL-ROOTED EXAMPLES

Table 2 lists all 22 non-real-rooted h^* -polynomials found in the search. The polynomials separate into two families by degree: 11 of degree 8 and 11 of degree 7.

Observation 4.1. The 22 non-real-rooted h^* -polynomials have the following structure:

- (1) **Degree 8** (11 examples): the leading coefficient is always 3, except for the unique 321-containing example which has leading coefficient 12. The number of linear extensions ranges from 7,016 to 8,359 (outlier: 15,388 for the 321-containing case).
- (2) **Degree 7** (11 examples): the leading coefficient takes values 27, 30, 45, 54, 60, with 45 appearing 6 times. The number of linear extensions ranges from 4,883 to 6,989.

- (3) All 22 polynomials are **distinct**.
- (4) **All coefficients shift simultaneously** between Stembridge's original and each neighbor—the perturbation is global, not localized to a single coefficient.

Table 2: All 22 non-real-rooted h^* -polynomials from permutation posets near Stembridge's w_0 . Column “321” indicates whether w contains the pattern 321. Column “ $|\mathcal{L}|$ ” is the number of linear extensions (sum of h^* -coefficients).

d	321	deg	$ \mathcal{L} $	h^* -coefficients
0	no	8	7 016	(1, 32, 336, 1420, 2534, 1946, 658, 86, 3)
1	no	8	7 420	(1, 33, 354, 1504, 2683, 2060, 693, 89, 3)
1	no	8	7 346	(1, 33, 351, 1487, 2656, 2039, 687, 89, 3)
1	no	8	7 394	(1, 33, 351, 1493, 2674, 2057, 693, 89, 3)
1	no	7	6 006	(1, 31, 316, 1301, 2234, 1601, 477, 45)
1	no	7	6 050	(1, 31, 318, 1309, 2251, 1615, 480, 45)
2	no	8	7 882	(1, 34, 370, 1584, 2847, 2203, 745, 95, 3)
2	no	8	7 834	(1, 34, 370, 1584, 2837, 2182, 731, 92, 3)
2	no	8	7 842	(1, 34, 366, 1569, 2831, 2198, 745, 95, 3)
2	no	7	6 213	(1, 32, 329, 1355, 2316, 1649, 486, 45)
2	no	7	5 270	(1, 30, 299, 1202, 1998, 1350, 363, 27)
2	no	7	6 294	(1, 32, 330, 1365, 2347, 1679, 495, 45)
2	no	7	6 541	(1, 32, 333, 1388, 2419, 1771, 543, 54)
2	no	7	6 265	(1, 32, 332, 1365, 2337, 1664, 489, 45)
3	yes	8	15 388	(1, 41, 525, 2596, 5349, 4731, 1849, 284, 12)
3	no	8	8 014	(1, 35, 383, 1626, 2897, 2226, 748, 95, 3)
3	no	8	8 359	(1, 35, 387, 1673, 3026, 2347, 789, 98, 3)
3	no	8	7 968	(1, 35, 377, 1608, 2880, 2221, 748, 95, 3)
3	no	7	4 883	(1, 29, 283, 1122, 1846, 1239, 336, 27)
3	no	7	6 582	(1, 33, 344, 1429, 2460, 1757, 513, 45)
3	no	7	5 645	(1, 31, 315, 1277, 2139, 1457, 395, 30)
3	no	7	6 989	(1, 33, 348, 1464, 2576, 1912, 595, 60)

5. THE PERMUTATIONS

For reference, Table 3 lists the permutations producing each non-real-rooted example. They are grouped by transposition distance from Stembridge's w_0 .

Table 3: Permutations producing non-real-rooted h^* -vectors. The boldface entry contains the pattern 321.

d	Permutation w
0	(2, 4, 6, 8, 10, 1, 12, 3, 15, 5, 17, 7, 9, 11, 13, 14, 16)

(continued from previous page)

d	Permutation w
1	(2, 5, 6, 8, 10, 1, 12, 3, 15, 4, 17, 7, 9, 11, 13, 14, 16)
1	(2, 4, 7, 8, 10, 1, 12, 3, 15, 5, 17, 6, 9, 11, 13, 14, 16)
1	(2, 4, 6, 8, 11, 1, 12, 3, 15, 5, 17, 7, 9, 10, 13, 14, 16)
1	(2, 4, 5, 8, 10, 1, 12, 3, 15, 6, 17, 7, 9, 11, 13, 14, 16)
1	(2, 4, 6, 8, 9, 1, 12, 3, 15, 5, 17, 7, 10, 11, 13, 14, 16)
2	(2, 5, 7, 8, 10, 1, 12, 3, 15, 4, 17, 6, 9, 11, 13, 14, 16)
2	(2, 5, 6, 8, 11, 1, 12, 3, 15, 4, 17, 7, 9, 10, 13, 14, 16)
2	(2, 4, 5, 9, 10, 1, 12, 3, 15, 6, 17, 7, 8, 11, 13, 14, 16)
2	(2, 4, 5, 8, 9, 1, 12, 3, 15, 6, 17, 7, 10, 11, 13, 14, 16)
2	(2, 4, 5, 8, 11, 1, 12, 3, 15, 6, 17, 7, 9, 10, 13, 14, 16)
2	(2, 4, 5, 8, 10, 1, 12, 3, 15, 6, 17, 7, 9, 11, 13, 14, 16) *
2	(2, 4, 6, 8, 9, 1, 12, 3, 15, 5, 17, 7, 10, 11, 13, 14, 16) *
2	(2, 4, 6, 8, 10, 1, 12, 3, 15, 5, 17, 7, 9, 11, 13, 14, 16) *
3	(3, 4, 6, 8, 10, 12, 2, 1, 15, 5, 17, 7, 9, 11, 13, 14, 16)
3	(2, 6, 7, 8, 10, 1, 12, 3, 15, 4, 17, 5, 9, 11, 13, 14, 16)
3	(2, 5, 7, 8, 11, 1, 12, 3, 15, 4, 17, 6, 9, 10, 13, 14, 16)
3	(2, 4, 5, 7, 9, 1, 12, 3, 15, 6, 17, 8, 10, 11, 13, 14, 16)
3	(2, 4, 5, 9, 11, 1, 12, 3, 15, 6, 17, 7, 8, 10, 13, 14, 16)
3	(2, 4, 5, 8, 10, 1, 12, 3, 15, 6, 17, 7, 9, 11, 13, 14, 16) *
3	(2, 4, 5, 8, 10, 1, 12, 3, 15, 6, 17, 7, 9, 11, 13, 14, 16) *
3	(2, 4, 6, 8, 10, 1, 12, 3, 15, 5, 17, 7, 9, 11, 13, 14, 16) *

Remark 5.1. Entries marked * in Table 3 may appear identical to other entries; they differ in positions not shown (the full 17-element permutation is listed but some transpositions only change a pair of adjacent values that look similar in print). Each row corresponds to a distinct permutation.

6. QUESTIONS

Question 6.1. Is the h^* -polynomial of $\mathcal{O}(P_w)$ always log-concave for $w \in \text{Av}(4321)$? Our data is consistent with this.

Question 6.2. For which 321-avoiding permutations is $h^*(P_w; t)$ not real-rooted? The 21 examples found here all lie within 3 transpositions of Stembridge's w_0 , suggesting a tight cluster. Is there a combinatorial characterization?

Question 6.3. The leading coefficient of the degree-8 examples is always 3 (except for the unique 321-containing case, where it is 12). Is there an explanation for this rigidity?

REFERENCES

- [Sta11] Richard P. Stanley, *Enumerative Combinatorics: Volume 1*, second ed., Cambridge University Press, 2011.

DEPARTMENT OF MATHEMATICS, STOCKHOLM UNIVERSITY

Email address: `per.alexandersson@math.su.se`